

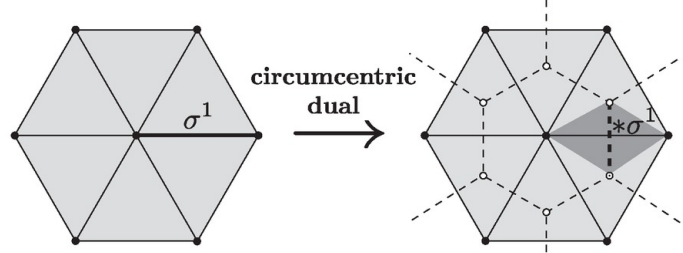
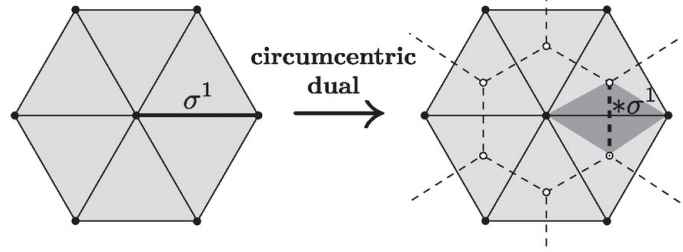
0707.4470: formula evidence

1 row shows a REFINEMENT below, not the OCR reading: 1 verified against the source. Each is marked [refined] beside its identifier. The model is unchanged — the original reading is still what ‘latex’ holds; this report chose the refinement for it. Rows: 0707.4470_FO0175

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0707.4470_F00001	1	■ 1.000	$\nabla \cdot \mathbf{D} = \rho$	$\nabla \cdot \mathbf{D} = \rho$	guishing attributes are that the Gauss constraint $\nabla \cdot \mathbf{D} = \rho$ is exactly conserved in
0707.4470_F00002	2	■ 1.000	$\mathbf{E} = -\nabla \phi$	$\mathbf{E} = -\nabla \phi$	a discrete sense, and electrostatic solutions of the form $\mathbf{E} = -\nabla \phi$ indeed remain
0707.4470_F00003	3	■ 0.677	$\mathbf{B} = \nabla \times \mathbf{A}$	$\mathbf{B} = \nabla \times \mathbf{A}$	sic relation $\mathbf{B} = \nabla \times \mathbf{A}$, where $\mathbf{B}$ is the magnetic flux and $\mathbf{A}$ is the magnetic vector
0707.4470_F00004	3	■ 0.677	$\mathbf{B}$	$\mathbf{B}$	sic relation $\mathbf{B} = \nabla \times \mathbf{A}$, where $\mathbf{B}$ is the magnetic flux and $\mathbf{A}$ is the magnetic vector
0707.4470_F00005	3	■ 1.000	$\nabla \cdot \nabla \times = 0$	$\nabla \cdot \nabla \times = 0$	potential. Because of the vector calculus identities $\nabla \cdot \nabla \times = 0$ and $\nabla \times \nabla = 0$, this
0707.4470_F00006	3	■ 1.000	$\nabla \times \nabla = 0$	$\nabla \times \nabla = 0$	potential. Because of the vector calculus identities $\nabla \cdot \nabla \times = 0$ and $\nabla \times \nabla = 0$, this
0707.4470_F00007	3	■ 0.959	$\mathbf{A} \mapsto \mathbf{A} + \nabla f$	$\mathbf{A} \mapsto \mathbf{A} + \nabla f$	cally divergence-free. Second, any transformation $\mathbf{A} \mapsto \mathbf{A} + \nabla f$ has no effect on $\mathbf{B}$;
0707.4470_F00008	3	■ 1.000	$\nabla \cdot \mathbf{D} - \rho$	$\nabla \cdot \mathbf{D} - \rho$	is $\nabla \cdot \mathbf{D} - \rho$ (which must be zero by Gauss’ law). A similar argument also explains
0707.4470_F00009	3	■ 1.000	ϕ	ϕ	under constant shifts in the scalar potential ϕ . Therefore, a proper variational
0707.4470_F00010	3	■ 1.000	$\mathrm{d}^2 = 0$	$\mathrm{d}^2 = 0$	tities such as $\mathrm{d}^2 = 0$ (which encapsulates the previous div-curl-grad relations)
0707.4470_F00011	4	■ 1.000	F	F	that are dual to one another (the spacetime forms F and $G = *F$), and hence they
0707.4470_F00012	4	■ 1.000	$G = * F$	$G = *F$	that are dual to one another (the spacetime forms F and $G = *F$), and hence they
0707.4470_F00013	5	■ 0.979	E	E	forms, we begin by replacing the electric field with a 1-form E and the magnetic
0707.4470_F00014	5	■ 0.995	B	B	flux density by a 2-form B . These have the coordinate expressions
0707.4470_F00015	6	■ 1.000	$\mathbf{E} = (E_x, E_y, E_z)$	$\mathbf{E} = (E_x, E_y, E_z)$	where $\mathbf{E} = (E_x, E_y, E_z)$ and $\mathbf{B} = (B_x, B_y, B_z)$. The motivation for choosing E as a
0707.4470_F00016	6	■ 1.000	$\mathbf{B} = (B_x, B_y, B_z)$	$\mathbf{B} = (B_x, B_y, B_z)$	where $\mathbf{E} = (E_x, E_y, E_z)$ and $\mathbf{B} = (B_x, B_y, B_z)$. The motivation for choosing E as a
0707.4470_F00017	6	■ 0.995	$\mathbf{E}$	$\mathbf{E}$	where $\mathbf{E}$ is integrated over curves and $\mathbf{B}$ is integrated over surfaces. Similarly,
0707.4470_F00018	6	■ 0.999	H	H	1-form H and a 2-form D .
0707.4470_F00019	6	■ 0.999	D	D	1-form H and a 2-form D .
0707.4470_F00020	6	■ 1.000	ϵ	ϵ	As shown in Bossavit and Kettunen (2000), we can view ϵ and μ as corresponding
0707.4470_F00021	6	■ 1.000	μ	μ	As shown in Bossavit and Kettunen (2000), we can view ϵ and μ as corresponding
0707.4470_F00022	6	■ 0.968	$*_{\epsilon}$	$*_{\epsilon}$	to Hodge operators $*_{\epsilon}$ and $*_{\mu}$, which map the 1-form “fields” to 2-form “fluxes” in
0707.4470_F00023	6	■ 0.968	$*_{\mu}$	$*_{\mu}$	to Hodge operators $*_{\epsilon}$ and $*_{\mu}$, which map the 1-form “fields” to 2-form “fluxes” in
0707.4470_F00024	6	■ 1.000	$\epsilon = \epsilon_0$	$\epsilon = \epsilon_0$	Note that in a vacuum, with $\epsilon = \epsilon_0$ and $\mu = \mu_0$ constant, one can simply express
0707.4470_F00025	6	■ 1.000	$\mu = \mu_0$	$\mu = \mu_0$	Note that in a vacuum, with $\epsilon = \epsilon_0$ and $\mu = \mu_0$ constant, one can simply express
0707.4470_F00026	6	■ 0.531	$\epsilon_0 = \mu_0 = c = 1$	$\epsilon_0 = \mu_0 = c = 1$	that $\epsilon_0 = \mu_0 = c = 1$, and hence ignoring the distinction between $\mathbf{E}$ and $\mathbf{D}$ and
0707.4470_F00027	6	■ 0.925	$\rho \, \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$	$\rho \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$	the <i>charge density 3-form</i> $\rho \, \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$, as well as the <i>current density 2-form</i>
0707.4470_F00028	6	■ 1.000	$J = J_x \, \mathrm{d}y \wedge \mathrm{d}z + J_y \, \mathrm{d}z \wedge \mathrm{d}x + J_z \, \mathrm{d}x \wedge \mathrm{d}y$	$J = J_x \, \mathrm{d}y \wedge \mathrm{d}z + J_y \, \mathrm{d}z \wedge \mathrm{d}x + J_z \, \mathrm{d}x \wedge \mathrm{d}y$	$J = J_x \, \mathrm{d}y \wedge \mathrm{d}z + J_y \, \mathrm{d}z \wedge \mathrm{d}x + J_z \, \mathrm{d}x \wedge \mathrm{d}y$. These are required to satisfy the continuity
0707.4470_F00029	6	■ 1.000	$\partial_t \rho + \mathrm{d}J = 0$	$\partial_t \rho + \mathrm{d}J = 0$	of charge condition $\partial_t \rho + \mathrm{d}J = 0$, which can be understood as a conservation law
0707.4470_F00030	7	■ 0.992	ρ	ρ	2.3. The Source 3-Form. Likewise, the charge density ρ and current density J
0707.4470_F00031	7	■ 0.992	J	J	2.3. The Source 3-Form. Likewise, the charge density ρ and current density J
0707.4470_F00032	7	■ 1.000	$\mathcal{J}$	$\mathcal{J}$	Having defined $\mathcal{J}$ in this way, the continuity of charge condition simply requires
0707.4470_F00033	7	■ 0.986	$\mathrm{d}\mathcal{J} = 0$	$\mathrm{d}\mathcal{J} = 0$	that $\mathcal{J}$ be closed, i.e., $\mathrm{d}\mathcal{J} = 0$.
0707.4470_F00034	7	■ 1.000	A	A	2.4. Electromagnetic Variational Principle. Let A be the electromagnetic po-
0707.4470_F00035	7	■ 1.000	$F = \mathrm{d}A$	$F = \mathrm{d}A$	tential 1-form, satisfying $F = \mathrm{d}A$, over the spacetime manifold X . Then define the
0707.4470_F00036	7	■ 1.000	X	X	tential 1-form, satisfying $F = \mathrm{d}A$, over the spacetime manifold X . Then define the
0707.4470_F00037	7	■ 1.000	α	α	Now, take a variation α of A , where α vanishes on the boundary ∂X . Then the

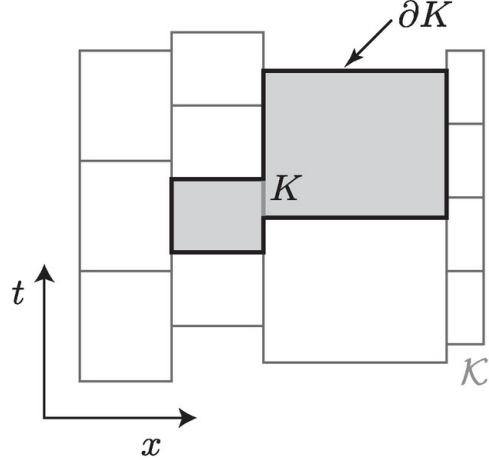
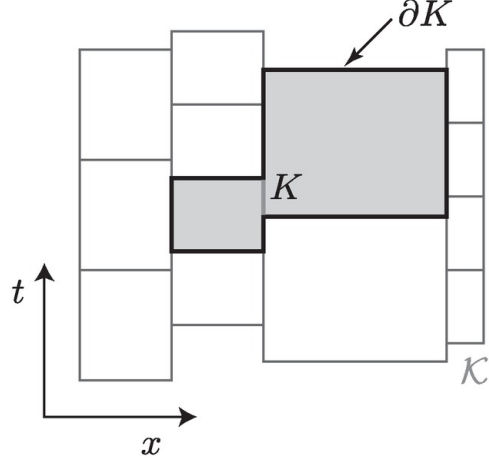
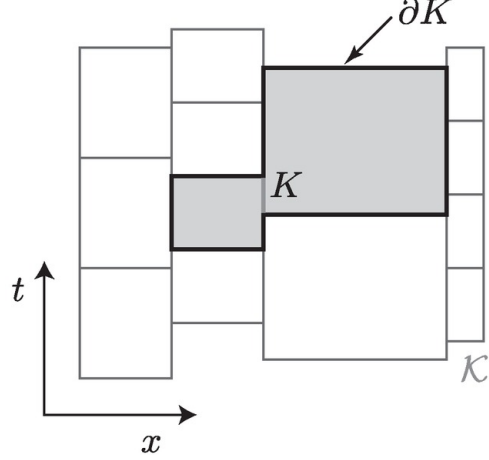
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0707.4470_F00038	7	■ 1.000	∂X	∂X	Now, take a variation α of A , where α vanishes on the boundary ∂X . Then the
0707.4470_F00039	7	■ 1.000	$G=*F=* \mathrm{d} A$	$G=*F=* \mathrm{d} A$	2.5. Variational Derivation of Maxwell's Equations. Since $G=*F=* \mathrm{d} A$, then
0707.4470_F00040	7	■ 0.999	$\mathrm{d} G=\mathscr{J}$	$\mathrm{d} G=\mathscr{J}$	clearly Equation 2.1 is equivalent to $\mathrm{d} G=\mathscr{J}$. Furthermore, since $\mathrm{d}^2=0$, it follows
0707.4470_F00041	8	■ 1.000	$\mathrm{d} F=\mathrm{d}^2 A=0$	$\mathrm{d} F=\mathrm{d}^2 A=0$	that $\mathrm{d} F=\mathrm{d}^2 A=0$. Hence, Maxwell's equations with respect to the Maxwell and
0707.4470_F00042	8	■ 0.991	(x, y, z, t)	(x, y, z, t)	Suppose now we choose the standard coordinate system (x, y, z, t) on Minkowski
0707.4470_F00043	8	■ 1.000	$X=\mathbb{R}^{3,1}$	$X=\mathbb{R}^{3,1}$	space $X=\mathbb{R}^{3,1}$, and define E and B through the relation $F=E \wedge \mathrm{d} t+B$. Then a
0707.4470_F00044	8	■ 1.000	$F=E \wedge \mathrm{d} t+B$	$F=E \wedge \mathrm{d} t+B$	space $X=\mathbb{R}^{3,1}$, and define E and B through the relation $F=E \wedge \mathrm{d} t+B$. Then a
0707.4470_F00045	8	■ 1.000	$G=*F=H \wedge \mathrm{d} t-D$	$G=*F=H \wedge \mathrm{d} t-D$	Likewise, if $\underline{G=*F=H \wedge \mathrm{d} t-D}$, then Equation 2.3 is equivalent to
0707.4470_F00046	8	■ 1.000	G	G	fields, while G acts as a Lagrange multiplier, weakly enforcing the constraint
0707.4470_F00047	8	■ 1.000	α, ϕ, γ	α, ϕ, γ	Then, taking the variation of the action along some α, ϕ, γ (vanishing on ∂X), we
0707.4470_F00048	9	■ 1.000	$\nabla \cdot \mathbf{B}$	$\nabla \cdot \mathbf{B}$	$\nabla \cdot \mathbf{B}$ and $\nabla \cdot \mathbf{D}-\rho$. Therefore, the divergence equations (2.5) and (2.7) can
0707.4470_F00049	9	■ 1.000	$\mathrm{d} A$	$\mathrm{d} A$	exterior derivative $\mathrm{d} A$ of the electromagnetic potential, and not on the value of
0707.4470_F00050	9	■ 1.000	$A \mapsto A+\mathrm{d} f$	$A \mapsto A+\mathrm{d} f$	$A \mapsto A+\mathrm{d} f$ leaves $\mathrm{d} A$, and hence Maxwell's equations, unchanged. Choosing
0707.4470_F00051	9	■ 1.000	$\phi=A(\partial / \partial t)=0$	$\phi=A(\partial / \partial t)=0$	potential $\phi=A(\partial / \partial t)=0$ (the so-called Weyl gauge or temporal gauge), and
0707.4470_F00052	9	■ 1.000	$\mathrm{d}^2 A=0$	$\mathrm{d}^2 A=0$	identity $\mathrm{d}^2 A=0$; they are not actually part of the Euler–Lagrange equations. A
0707.4470_F00053	11	■ 1.000	n	n	n -dimensional manifold X is replaced by a discrete mesh—precisely, by a cell
0707.4470_F00054	11	■ 1.000	K	K	2-dimensional surface. We will generally denote the complex by K , and a cell in
0707.4470_F00055	11	■ 1.000	σ	σ	the complex by σ .
0707.4470_F00056	11	■ 0.999	$* K$	$* K$	Given a mesh K , one can construct a <i>dual mesh</i> $* K$, where each k -cell σ
0707.4470_F00057	11	■ 0.999	k	k	Given a mesh K , one can construct a <i>dual mesh</i> $* K$, where each k -cell σ
0707.4470_F00058	11	■ 0.364	$(n-k)$	$(n-k)$	corresponds to a dual $(n-k)$ -cell $* \sigma$. ($* K$ is “dual” to K in the sense of a graph
0707.4470_F00059	11	■ 0.364	$* \sigma$	$* \sigma$	corresponds to a dual $(n-k)$ -cell $* \sigma$. ($* K$ is “dual” to K in the sense of a graph
0707.4470_F00060	11	■ 0.364	$(* K$	$(* K$	corresponds to a dual $(n-k)$ -cell $* \sigma$. ($* K$ is “dual” to K in the sense of a graph
0707.4470_F00061	11	■ 0.972	$(n-1)$	$(n-1)$	corresponding n -simplices share an $(n-1)$ -simplex, and so on. This is called
0707.4470_F00062	11	■ 1.000	α^k	α^k	ferential forms. A discrete k -form α^k assigns a real number to each oriented
0707.4470_F00063	11	■ 1.000	σ^k	σ^k	k -dimensional cell σ^k in the mesh K . (The superscripts k are not actually re-
0707.4470_F00064	11	■ 1.000	$\left\langle \alpha^k, \sigma^k \right\rangle$	$\left\langle \alpha^k, \sigma^k \right\rangle$	form or cell we are dealing with.) This value is denoted by $\left\langle \alpha^k, \sigma^k \right\rangle$, and can be
0707.4470_F00065	12	—	σ^1	σ^1	
0707.4470_F00066	12	—	$* \sigma^1$	$* \sigma^1$	

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0707.4470_F00067	12	—	$\mathrm{CH}\left(\sigma^1, * \sigma^1\right)$	$\mathrm{CH}\left(\sigma^1, *\sigma^1\right)$	
0707.4470_F00068	12	■ 1.000	$\angle\cdot, \cdot\angle$	$\langle\cdot, \cdot\rangle$	are called <i>chains</i> , and discrete differential forms are <i>cochains</i> , where $\langle\cdot, \cdot\rangle$ is the
0707.4470_F00069	12	■ 0.994	$n-k$	$n-k$	a natural correspondence between primal k -forms and dual $(n-k)$ -forms, since
0707.4470_F00070	12	■ 0.884	$k+1$	$k+1$	Therefore, if α is a discrete differential k -form, then the $(k+1)$ -form $d\alpha$ is defined
0707.4470_F00071	12	■ 0.884	$\mathrm{d} \alpha$	$d\alpha$	Therefore, if α is a discrete differential k -form, then the $(k+1)$ -form $d\alpha$ is defined
0707.4470_F00072	12	■ 0.604	$(k+1)$	$(k+1)$	on any $(k+1)$ -chain σ by
0707.4470_F00073	12	■ 1.000	$\partial \sigma$	$\partial\sigma$	where $\partial\sigma$ is the k -chain boundary of σ . For this reason, d is often called the
0707.4470_F00074	13	■ 1.000	$* \alpha$	$*\alpha$	versions can be substituted. Given a discrete form α , its <i>Hodge star</i> $*\alpha$ is defined
0707.4470_F00075	13	■ 1.000	$ \sigma $	$ \sigma $	where $ \sigma $ and $ \sigma $ are the volumes of these elements, and κ is the causality
0707.4470_F00076	13	■ 1.000	$ \sigma $	$ \sigma $	where $ \sigma $ and $ \sigma $ are the volumes of these elements, and κ is the causality
0707.4470_F00077	13	■ 1.000	κ	κ	where $ \sigma $ and $ \sigma $ are the volumes of these elements, and κ is the causality
0707.4470_F00078	13	■ 0.755	$\mathbb{L}^2$	$\mathbb{L}^2$	<i>Inner Product</i> . Define the $\mathbb{L}^2$ inner product $(\cdot, \cdot)$ between two primal k -forms to
0707.4470_F00079	13	■ 0.755	$(\cdot, \cdot)$	$(\cdot, \cdot)$	<i>Inner Product</i> . Define the $\mathbb{L}^2$ inner product $(\cdot, \cdot)$ between two primal k -forms to
0707.4470_F00080	13	■ 0.978	$\mathrm{CH}(\sigma, *\sigma)$	$\mathrm{CH}(\sigma, *\sigma)$	where the sum is taken over all k -dimensional elements σ , and $\mathrm{CH}(\sigma, *\sigma)$ is
0707.4470_F00081	13	■ 1.000	$\sigma \cup *\sigma$	$\sigma \cup *\sigma$	the n -dimensional convex hull of $\sigma \cup *\sigma$ (see Figure 1). The final equality holds
0707.4470_F00082	13	■ 0.991	$ \mathrm{CH}(\sigma, *\sigma) = \binom{n}{k}^{-1} \sigma \sigma $	$ \mathrm{CH}(\sigma, *\sigma) = \binom{n}{k}^{-1} \sigma \sigma $	one another, and hence $ \mathrm{CH}(\sigma, *\sigma) = \binom{n}{k}^{-1} \sigma \sigma $. (Indeed, this is one of the
0707.4470_F00083	13	■ 1.000	$\alpha \wedge *\beta$	$\alpha \wedge *\beta$	primal-dual convex hulls.) This inner product can be expressed in terms of $\alpha \wedge *\beta$,
0707.4470_F00084	13	■ 0.865	δ	δ	d and $*$, we immediately have a discrete codifferential δ , with the same formal
0707.4470_F00085	13	■ 0.995	$\mathrm{d}, *, \delta$	$d, *, \delta$	dual discrete forms, along with the corresponding operators $d, *, \delta$, for the case
0707.4470_F00086	13	■ 1.000	σ_i^k	σ_i^k	on each k -cell of the mesh. That is, given a list of k -cells σ_i^k , the entries of the
0707.4470_F00087	13	■ 0.803	$\alpha_i = \langle \alpha, \sigma_i^k \rangle$	$\alpha_i = \langle \alpha, \sigma_i^k \rangle$	vector are $\alpha_i = \langle \alpha, \sigma_i^k \rangle$. The exterior derivative d , taking k -forms to $(k+1)$ -forms,
0707.4470_F00088	14	—	$k=0, 1, 2, 3$	$k=0, 1, 2, 3$	

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0707.4470_F00089	14	—	*	*	
0707.4470_F00090	14	—	$*^{-1}$	$*^{-1}$	
0707.4470_F00091	14	■ 0.510	$\kappa\left(\sigma_i^k\right) \frac{\left \sigma_i^k\right }{\left \sigma_i^k\right }$	$\kappa\left(\sigma_i^k\right) \frac{\left \sigma_i^k\right }{\left \sigma_i^k\right }$	of the diagonal Hodge star, it is the diagonal matrix with entries $\kappa\left(\sigma_i^k\right) \frac{\left \sigma_i^k\right }{\left \sigma_i^k\right }$. The
0707.4470_F00092	14	■ 1.000	∂K	∂K	∂K . For example, in electromagnetism, we may wish to set initial conditions for
0707.4470_F00093	14	■ 1.000	$t_{\{0\}}$	t_0	E and B at time t_0 — but while B is defined on ∂K at t_0 , E is not. In fact, as we
0707.4470_F00094	14	■ 1.000	$t_{\{0\}}, E$	t_0, E	E and B at time t_0 — but while B is defined on ∂K at t_0 , E is not. In fact, as we
0707.4470_F00095	14	■ 1.000	$t_{\{1\}}$	t_1	will see, E lives on edges that are extruded between the time slices t_0 and t_1 , so
0707.4470_F00096	14	■ 1.000	$t_{\{1 / 2\}}$	$t_{1 / 2}$	unless we modify our definitions, we can only initialize E at the half-step $t_{1 / 2}$.
0707.4470_F00097	15	■ 0.608	$n-1$	$n-1$	addition of a dual vertex at the circumcenter of each boundary $(n-1)$ -simplex, in
0707.4470_F00098	15	■ 0.844	$(k-1)$	$(k-1)$	dimensional shell. That is, each boundary $(k-1)$ -simplex can be thought of
0707.4470_F00099	15	■ 0.997	∂K	∂K	∂K . In fact, the boundary of the dual is now equal to the dual of the boundary,
0707.4470_F00100	15	■ 1.000	$\partial(*K)=\partial K$	$\partial(*K)=\partial K$	i.e., $\partial(*K)=\partial K$. Returning to the example of initial conditions on E and B , we
0707.4470_F00101	15	■ 1.000	$t_{-\epsilon}$	$t_{-\epsilon}$	vanishingly thin faces (i.e., extruded between some $t_{-\epsilon}$ and t_0 for $\epsilon \rightarrow 0$). Notice
0707.4470_F00102	15	■ 1.000	$\epsilon \rightarrow 0$	$\epsilon \rightarrow 0$	vanishingly thin faces (i.e., extruded between some $t_{-\epsilon}$ and t_0 for $\epsilon \rightarrow 0$). Notice
0707.4470_F00103	16	■ 0.862	$k-1$	$k-1$	boundary terms. If α is a primal $(k-1)$ -form and β is a primal k -form, then
0707.4470_F00104	16	■ 0.862	β	β	boundary terms. If α is a primal $(k-1)$ -form and β is a primal k -form, then
0707.4470_F00105	16	■ 0.894	$* \beta$	$* \beta$	In the boundary integral, α is still a primal $(k-1)$ -form on ∂K , while $* \beta$ is an
0707.4470_F00106	16	■ 0.991	F, G	F, G	First, we will find a sensible way to define the discrete forms F, G , and $\mathcal{J}$ on a
0707.4470_F00107	16	■ 0.989	$\mathbb{R}^{3,1}$	$\mathbb{R}^{3,1}$	generalized spacetime meshes, e.g., an arbitrary meshing of $\mathbb{R}^{3,1}$ by 4-simplices.
0707.4470_F00108	16	■ 1.000	p	p	more abstract perspective of higher-level “ p -form” versions of electromagnetism
0707.4470_F00109	16	■ 1.000	$\Delta x, \Delta y, \Delta z, \Delta t$	$\Delta x, \Delta y, \Delta z, \Delta t$	let us also suppose that the grid has uniform space and time steps $\Delta x, \Delta y, \Delta z, \Delta t$.
0707.4470_F00110	17	■ 0.925	$E_x \Delta x \Delta t$	$E_x \Delta x \Delta t$	<i>six types of 2-faces in a 4-D rectangular grid</i> . Simply assign the values $E_x \Delta x \Delta t$
0707.4470_F00111	17	■ 1.000	$x t$	$x t$	to faces parallel to the $x t$ -plane, $E_y \Delta y \Delta t$ to faces parallel to the $y t$ -plane, and
0707.4470_F00112	17	■ 1.000	$E_y \Delta y \Delta t$	$E_y \Delta y \Delta t$	to faces parallel to the $x t$ -plane, $E_y \Delta y \Delta t$ to faces parallel to the $y t$ -plane, and
0707.4470_F00113	17	■ 1.000	$y t$	$y t$	to faces parallel to the $x t$ -plane, $E_y \Delta y \Delta t$ to faces parallel to the $y t$ -plane, and
0707.4470_F00114	17	■ 1.000	$E_z \Delta z \Delta t$	$E_z \Delta z \Delta t$	$E_z \Delta z \Delta t$ to faces parallel to the $z t$ -plane. Likewise, assign $B_x \Delta y \Delta z$ to faces
0707.4470_F00115	17	■ 1.000	$z t$	$z t$	$E_z \Delta z \Delta t$ to faces parallel to the $z t$ -plane. Likewise, assign $B_x \Delta y \Delta z$ to faces

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0707.4470_F00116	17	■ 1.000	$B_{\{x\}} \Delta y \Delta z$	$B_x \Delta y \Delta z$	$E_z \Delta z \Delta t$ to faces parallel to the zt -plane. Likewise, assign $B_x \Delta y \Delta z$ to faces
0707.4470_F00117	17	■ 1.000	$y z$	yz	parallel to the yz -plane, $B_y \Delta z \Delta x$ to faces parallel to the xz -plane, and $B_z \Delta x \Delta y$
0707.4470_F00118	17	■ 1.000	$B_{\{y\}} \Delta z \Delta x$	$B_y \Delta z \Delta x$	parallel to the yz -plane, $B_y \Delta z \Delta x$ to faces parallel to the xz -plane, and $B_z \Delta x \Delta y$
0707.4470_F00119	17	■ 1.000	$x z$	xz	parallel to the yz -plane, $B_y \Delta z \Delta x$ to faces parallel to the xz -plane, and $B_z \Delta x \Delta y$
0707.4470_F00120	17	■ 1.000	$B_{\{z\}} \Delta x \Delta y$	$B_z \Delta x \Delta y$	parallel to the yz -plane, $B_y \Delta z \Delta x$ to faces parallel to the xz -plane, and $B_z \Delta x \Delta y$
0707.4470_F00121	17	■ 0.998	$x y$	xy	to faces parallel to the xy -plane. This is pictured in Figure 4.
0707.4470_F00122	17	■ 0.930	$\left[x_{\{k\}}, x_{\{k+1\}}\right] \times$	$[x_k, x_{k+1}] \times$	Let us look at these values on the faces of a typical 4-rectangle $[x_k, x_{k+1}] \times$
0707.4470_F00123	17	■ 1.000	$\left[y_{\{l\}}, y_{\{l+1\}}\right] \times \left[z_{\{m\}}, z_{\{m+1\}}\right] \times \left[t_{\{n\}}, t_{\{n+1\}}\right]$	$[y_l, y_{l+1}] \times [z_m, z_{m+1}] \times [t_n, t_{n+1}]$	$[y_l, y_{l+1}] \times [z_m, z_{m+1}] \times [t_n, t_{n+1}]$. To simplify the notation, we can index each
0707.4470_F00124	17	■ 0.965	$\left.F\right _{\left\{k+\frac{1}{2}\right\}, l, m}^{n+\frac{1}{2}}$	$F _{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}}$	value of F by the midpoint of the 2-face on which it lives: for example, $F _{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}}$
0707.4470_F00125	17	■ 1.000	$\left[x_{\{k\}}, x_{\{k+1\}}\right] \times \left[y_{\{l\}}\right] \times \left[z_{\{m\}}\right] \times \left[t_{\{n\}}, t_{\{n+1\}}\right]$	$[x_k, x_{k+1}] \times \{y_l\} \times \{z_m\} \times [t_n, t_{n+1}]$	is stored on the face $[x_k, x_{k+1}] \times \{y_l\} \times \{z_m\} \times [t_n, t_{n+1}]$, parallel to the xt -plane.
0707.4470_F00126	18	■ 1.000	$\mathrm{d} F$	dF	at the DEC interpretation of dF . Since dF is a discrete 3-form, it takes values on
0707.4470_F00127	20	■ 1.000	$\mathbf{D}=\epsilon \mathbf{E}, \mathbf{B}=\mu \mathbf{H}$	$\mathbf{D} = \epsilon \mathbf{E}, \mathbf{B} = \mu \mathbf{H}$	tion 5.2 for details) and making the substitutions $\mathbf{D} = \epsilon \mathbf{E}, \mathbf{B} = \mu \mathbf{H}$, the remaining
0707.4470_F00128	21	■ 1.000	$\kappa=1$	$\kappa = 1$	the value $\kappa = 1$. Second, there are rectangular faces that live between time steps.
0707.4470_F00129	21	■ 1.000	$\kappa=-1$	$\kappa = -1$	they have one timelike edge, these faces satisfy $\kappa = -1$. Again, the circumcentric-
0707.4470_F00130	21	■ 1.000	$E \Delta t$	$E \Delta t$	Therefore, let us assign B to the purely spacelike faces and $E \Delta t$ to the mixed
0707.4470_F00131	21	■ 1.000	$H \Delta t$	$H \Delta t$	$H \Delta t$ and spacelike dual faces should store D ; see Figure 6.
0707.4470_F00132	21	■ 1.000	$B^{\{n\}}$	B^n	Section 3.2. This leads to the arrays B^n and H^n at whole time steps n , and $E^{n+1/2}$
0707.4470_F00133	21	■ 1.000	$H^{\{n\}}$	H^n	Section 3.2. This leads to the arrays B^n and H^n at whole time steps n , and $E^{n+1/2}$
0707.4470_F00134	21	■ 1.000	$E^{\{n+1 / 2\}}$	$E^{n+1/2}$	Section 3.2. This leads to the arrays B^n and H^n at whole time steps n , and $E^{n+1/2}$
0707.4470_F00135	21	■ 1.000	$D^{\{n+1 / 2\}}$	$D^{n+1/2}$	and $D^{n+1/2}$ at half time steps. Let d_1 denote the edges-to-faces incidence matrix
0707.4470_F00136	21	■ 1.000	d_1	d_1	and $D^{n+1/2}$ at half time steps. Let d_1 denote the edges-to-faces incidence matrix
0707.4470_F00137	21	■ 1.000	d_1^T	d_1^T	Similarly, the transpose d_1^T corresponds to the exterior derivative from spatial
0707.4470_F00138	21	■ 1.000	$\mathrm{d} F=0$	$dF = 0$	dual 1-forms to dual 2-forms. Then the equation $dF = 0$, evaluated on all prisma
0707.4470_F00139	23	■ 1.000	$\Delta t_{\sigma}=t_{\sigma}^{n+1}-t_{\sigma}^n$	$\Delta t_{\sigma} = t_{\sigma}^{n+1} - t_{\sigma}^n$	For example, one might assign each face a fixed time step size $\Delta t_{\sigma} = t_{\sigma}^{n+1} - t_{\sigma}^n$,
0707.4470_F00140	23	■ 1.000	Δt	Δt	taking equal time steps <i>within</i> each element, but with Δt varying <i>across</i> elements.
0707.4470_F00141	23	■ 1.000	$\Theta_{\sigma} \cap \Theta_{\sigma'}=\left\{t_0\right\}$	$\Theta_{\sigma} \cap \Theta_{\sigma'} = \{t_0\}$	no two faces take the same time step: that is, $\Theta_{\sigma} \cap \Theta_{\sigma'} = \{t_0\}$ for $\sigma \neq \sigma'$.
0707.4470_F00142	23	■ 1.000	$\sigma \neq \sigma'$	$\sigma \neq \sigma'$	no two faces take the same time step: that is, $\Theta_{\sigma} \cap \Theta_{\sigma'} = \{t_0\}$ for $\sigma \neq \sigma'$.
0707.4470_F00143	23	■ 1.000	e	e	In order to keep proper time at the edges e where multiple faces with different
0707.4470_F00144	23	■ 1.000	$t_{\{e\}^{\{k+1 / 2\}}}=\left(t_{\{e\}^{\{k+1\}}}+t_{\{e\}^{\{k\}}}\right) / 2$	$t_e^{k+1/2} = (t_e^{k+1} + t_e^k) / 2$	where $t_e^{k+1/2} = (t_e^{k+1} + t_e^k) / 2$. The values stored on a primal AVI mesh are shown
0707.4470_F00145	23	■ 1.000	$\Theta_{\{e\}} \supset \Theta_{\sigma}$	$\Theta_e \supset \Theta_{\sigma}$	Since $\Theta_e \supset \Theta_{\sigma}$ when $e \subset \sigma$, each spatial edge e takes more time steps than any
0707.4470_F00146	23	■ 1.000	$e \subset \sigma$	$e \subset \sigma$	Since $\Theta_e \supset \Theta_{\sigma}$ when $e \subset \sigma$, each spatial edge e takes more time steps than any
0707.4470_F00147	24	■ 1.000	$\mathbb{K}_{\left\{t_{\sigma}^n=t_e^m\right\}}$	$\mathbb{K}_{\{t_{\sigma}^n=t_e^m\}}$	where $\mathbb{K}_{\{t_{\sigma}^n=t_e^m\}}$ equals 1 when face σ has $t_{\sigma}^n = t_e^m$ for some n , and 0 otherwise.
0707.4470_F00148	24	■ 1.000	$t_{\sigma}^n=t_e^m$	$t_{\sigma}^n = t_e^m$	where $\mathbb{K}_{\{t_{\sigma}^n=t_e^m\}}$ equals 1 when face σ has $t_{\sigma}^n = t_e^m$ for some n , and 0 otherwise.
0707.4470_F00149	24	■ 1.000	t_{σ}^{n+1}	t_{σ}^{n+1}	(1) Pick the minimum time t_{σ}^{n+1} where B_{σ}^{n+1} has not yet been computed.
0707.4470_F00150	24	■ 1.000	B_{σ}^{n+1}	B_{σ}^{n+1}	(1) Pick the minimum time t_{σ}^{n+1} where B_{σ}^{n+1} has not yet been computed.
0707.4470_F00151	24	■ 1.000	$H_{\sigma}^{n+1}=\mu^{-1} B_{\sigma}^{n+1}$	$H_{\sigma}^{n+1} = \mu^{-1} B_{\sigma}^{n+1}$	(3) Update $H_{\sigma}^{n+1} = \mu^{-1} B_{\sigma}^{n+1}$.
0707.4470_F00152	24	■ 1.000	$D_e^{m+3 / 2}$	$D_e^{m+3/2}$	(4) Advance $D_e^{m+3/2}$ on neighboring edges $e \subset \sigma$ according to Equation 4.4.
0707.4470_F00153	24	■ 1.000	$E_e^{m+3 / 2}=\epsilon^{-1} D_e^{m+3 / 2}$	$E_e^{m+3/2} = \epsilon^{-1} D_e^{m+3/2}$	(5) Update $E_e^{m+3/2} = \epsilon^{-1} D_e^{m+3/2}$.
0707.4470_F00154	24	■ 0.902	$\mathbf{x}$	$\mathbf{x}$	pared to the AVI for elasticity, A plays the role of the positions $\mathbf{x}$, while E plays
0707.4470_F00155	24	■ 1.000	$\dot{\mathbf{x}}$	$\dot{\mathbf{x}}$	the role of the (negative) velocities $\dot{\mathbf{x}}$. The algorithm is given as pseudocode in
0707.4470_F00156	25	■ 0.748	$A_{\{e\}} \leftarrow A_{\{e\}}^0, E_{\{e\}} \leftarrow E_{\{e\}}^{1 / 2}, \tau_{\{e\}} \leftarrow t_{\{0\}} / /$	$A_e \leftarrow A_e^0, E_e \leftarrow E_e^{1/2}, \tau_e \leftarrow t_0 //$	$A_e \leftarrow A_e^0, E_e \leftarrow E_e^{1/2}, \tau_e \leftarrow t_0 //$ Store initial field values and times

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0707.4470_F00157	25	■ 1.000	$\tau_{\sigma} \leftarrow t_0$	$\tau_{\sigma} \leftarrow t_0$	$\tau_{\sigma} \leftarrow t_0$
0707.4470_F00158	25	■ 0.999	t_{σ}^1	t_{σ}^1	Compute the next update time t_{σ}^1
0707.4470_F00159	25	■ 0.851	$Q . \mathrm{push}(\left(t_{\sigma}^1, \sigma\right), \sigma) / /$	$Q . \mathrm{push}\left(t_{\sigma}^1, \sigma\right) / /$	$Q . \mathrm{push}\left(t_{\sigma}^1, \sigma\right) / /$ Push element onto queue with its next update time
0707.4470_F00160	25	■ 0.873	$(t, \sigma) \leftarrow Q . \operatorname{pop}() / /$	$(t, \sigma) \leftarrow Q . \mathrm{pop}() / /$	$(t, \sigma) \leftarrow Q . \mathrm{pop}() / /$ Pop next element σ and time t from queue
0707.4470_F00161	25	■ 0.873	t	t	$(t, \sigma) \leftarrow Q . \mathrm{pop}() / /$ Pop next element σ and time t from queue
0707.4470_F00162	25	■ 1.000	$A_{\{e\}} \leftarrow A_{\{e\}} - E_{\{e\}} \left(t - \tau_{\{e\}}\right) / /$	$A_e \leftarrow A_e - E_e\left(t - \tau_e\right) / /$	$A_e \leftarrow A_e - E_e\left(t - \tau_e\right) / /$ Update neighboring values of A at time t
0707.4470_F00163	25	■ 0.570	$t <$	$t <$	if $t <$ final-time then
0707.4470_F00164	25	■ 1.000	$B_{\sigma} \leftarrow \mathrm{d}_1 A_e$	$B_{\sigma} \leftarrow \mathrm{d}_1 A_e$	$B_{\sigma} \leftarrow \mathrm{d}_1 A_e$
0707.4470_F00165	25	■ 1.000	$H_{\sigma} \leftarrow *_\mu B_{\sigma}$	$H_{\sigma} \leftarrow *_\mu B_{\sigma}$	$H_{\sigma} \leftarrow *_\mu B_{\sigma}$
0707.4470_F00166	25	■ 0.998	$D_e \leftarrow *_\epsilon E_e$	$D_e \leftarrow *_\epsilon E_e$	$D_e \leftarrow *_\epsilon E_e$
0707.4470_F00167	25	■ 1.000	$D_{\{e\}} \leftarrow D_{\{e\}} + \mathrm{d}_1(e, \sigma) H_{\sigma} \left(t - \tau_{\sigma}\right) \mathrm{d}_1(e, \sigma) H_{\sigma}\left(t - \tau_{\sigma}\right)$	$D_e \leftarrow D_e + \mathrm{d}_1(e, \sigma) H_{\sigma}\left(t - \tau_{\sigma}\right)$	$D_e \leftarrow D_e + \mathrm{d}_1(e, \sigma) H_{\sigma}\left(t - \tau_{\sigma}\right)$
0707.4470_F00168	25	■ 1.000	$E_e \leftarrow *_\epsilon D_e$	$E_e \leftarrow *_\epsilon D_e$	$E_e \leftarrow *_\epsilon D_e$
0707.4470_F00169	25	■ 0.999	$\tau_{\sigma} \leftarrow t / /$	$\tau_{\sigma} \leftarrow t / /$	$\tau_{\sigma} \leftarrow t / /$ Update element's time
0707.4470_F00170	25	■ 0.997	t_{σ}^{next}	t_{σ}^{next}	Compute the next update time t_{σ}^{next}
0707.4470_F00171	25	■ 0.438	$Q . \mathrm{push}(\left(t_{\sigma}^{\text{next}}, \sigma\right), \sigma) / /$	$Q . \mathrm{push}\left(t_{\sigma}^{\text{next}}, \sigma\right) / /$	$Q . \mathrm{push}\left(t_{\sigma}^{\text{next}}, \sigma\right) / /$ Schedule σ for next update
0707.4470_F00172	25	■ 1.000	x	x	domly partitioned the x - and y -axes, so that each element has completely unique
0707.4470_F00173	25	■ 1.000	y	y	domly partitioned the x - and y -axes, so that each element has completely unique
0707.4470_F00174	27	■ 0.998	$3+1$	$3+1$	so a 3+1 mesh (as in the previous subsections) is almost always adequate, and
0707.4470_F00175 [refined: source]	29	■ 0.919	$\mathscr{J}$	$\mathscr{J}$	Given a discrete 1-form A and dual source 3-form $\mathscr{J}$, define the discrete
0707.4470_F00176	29	■ 1.000	$\mathrm{d} * \mathrm{d} A = \mathscr{J}$	$\mathrm{d} * \mathrm{d} A = \mathscr{J}$	Euler-Lagrange equations are therefore $\mathrm{d} * \mathrm{d} A = \mathscr{J}$. Defining the discrete 2-forms
0707.4470_F00177	29	—	$\square$	$\square$	$\square$
0707.4470_F00178	29	■ 1.000	$\theta_{\mathscr{L}}$	$\theta_{\mathscr{L}}$	to vanish at the boundary. As originally presented, the Cartan form $\theta_{\mathscr{L}}$ is an
0707.4470_F00179	29	■ 1.000	$(n+1)$	$(n+1)$	$(n+1)$ -form, where the n -dimensional boundary integral is then obtained by
0707.4470_F00180	29	■ 0.976	$(n+2)$	$(n+2)$	contracting $\theta_{\mathscr{L}}$ with a variation. The multisymplectic $(n+2)$ -form $\omega_{\mathscr{L}}$ is then
0707.4470_F00181	29	■ 0.976	$\omega_{\mathscr{L}}$	$\omega_{\mathscr{L}}$	contracting $\theta_{\mathscr{L}}$ with a variation. The multisymplectic $(n+2)$ -form $\omega_{\mathscr{L}}$ is then
0707.4470_F00182	29	■ 1.000	$\omega_{\mathscr{L}} = -\mathrm{d}\theta_{\mathscr{L}}$	$\omega_{\mathscr{L}} = -\mathrm{d}\theta_{\mathscr{L}}$	given by $\omega_{\mathscr{L}} = -\mathrm{d}\theta_{\mathscr{L}}$. Contracting $\omega_{\mathscr{L}}$ with two arbitrary variations gives an
0707.4470_F00183	29	■ 1.000	$\mathrm{d}^2 = 0$	$\mathrm{d}^2 = 0$	<i>multisymplectic form formula</i> , which results from the identity $\mathrm{d}^2 = 0$. In the
0707.4470_F00184	29	■ 1.000	$n=0$	$n=0$	special case of mechanics, where $n=0$, the boundary consists of the initial and
0707.4470_F00185	29	■ 1.000	ω_L	ω_L	ω_L is preserved by the time flow.
0707.4470_F00186	30	■ 1.000	$K \subset \mathscr{K}$	$K \subset \mathscr{K}$	<i>Proof.</i> Let $K \subset \mathscr{K}$ be an arbitrary subcomplex, and consider the discrete action
0707.4470_F00187	30	■ 1.000	S_d	S_d	functional S_d restricted to K . Suppose now that we take a discrete variation α ,
0707.4470_F00188	30	■ 1.000	$\theta_{\mathscr{L}_d}$	$\theta_{\mathscr{L}_d}$	Then we can define the <i>Cartan form</i> $\theta_{\mathscr{L}_d}$ by
0707.4470_F00189	30	■ 1.000	$\omega_{\mathscr{L}_d} = -\mathrm{d}\theta_{\mathscr{L}_d}$	$\omega_{\mathscr{L}_d} = -\mathrm{d}\theta_{\mathscr{L}_d}$	Finally, defining the multisymplectic form $\omega_{\mathscr{L}_d} = -\mathrm{d}\theta_{\mathscr{L}_d}$, and using the fact that
0707.4470_F00190	30	■ 1.000	$\mathrm{d}^2 S_d = 0$	$\mathrm{d}^2 S_d = 0$	$\mathrm{d}^2 S_d = 0$, we get the relation
0707.4470_F00191	30	■ 1.000	α, β	α, β	for all variations α, β ; Equation 5.2 is a discrete version of the multisymplec-

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0707.4470_F00192	31	—	K	$\mathcal{K}$	
0707.4470_F00193	31	—	ω_{L_d}	$\omega_{\mathcal{L}_d}$	
0707.4470_F00194	31	—	$\omega_{\mathrm{L}_d} \cdot \alpha \cdot \beta$	$\omega_{\mathcal{L}_d} \cdot \alpha \cdot \beta$	
0707.4470_F00195	31	■ 1.000	$\rho = \nabla \cdot \mathbf{D}$	$\rho = \nabla \cdot \mathbf{D}$	turns out to be the charge density $\rho = \nabla \cdot \mathbf{D}$, which justifies its elimination from
0707.4470_F00196	31	■ 1.000	f	f	are invariant under gauge transformations of the form $A \mapsto A + df$, where f is any
0707.4470_F00197	31	■ 1.000	$A_t = 0$	$A_t = 0$	Weyl gauge, so that the time component $A_t = 0$. Therefore, we can assume that
0707.4470_F00198	31	■ 0.749	A_x, A_y, A_z	A_x, A_y, A_z	In fact, A_x, A_y, A_z are precisely the components of the familiar vector potential $\mathbf{A}$,
0707.4470_F00199	31	■ 0.749	$\mathbf{A}$	$\mathbf{A}$	In fact, A_x, A_y, A_z are precisely the components of the familiar vector potential $\mathbf{A}$,
0707.4470_F00200	31	■ 0.880	$A = \mathbf{A}^b$	$A = \mathbf{A}^b$	i.e., $A = \mathbf{A}^b$.
0707.4470_F00201	31	■ 1.000	d_t	d_t	nate, we can now define two new “partial exterior derivative” operators, d_t (time)
0707.4470_F00202	32	■ 0.934	d_s	d_s	and d_s (space), where $\mathrm{d} = \mathrm{d}_t + \mathrm{d}_s$. Since A contains no dt terms, $\mathrm{d}_s A$ is a 2-form
0707.4470_F00203	32	■ 0.934	$\mathrm{d} = \mathrm{d}_t + \mathrm{d}_s$	$\mathrm{d} = \mathrm{d}_t + \mathrm{d}_s$	and d_s (space), where $\mathrm{d} = \mathrm{d}_t + \mathrm{d}_s$. Since A contains no dt terms, $\mathrm{d}_s A$ is a 2-form
0707.4470_F00204	32	■ 0.934	d_t	d_t	and d_s (space), where $\mathrm{d} = \mathrm{d}_t + \mathrm{d}_s$. Since A contains no dt terms, $\mathrm{d}_s A$ is a 2-form
0707.4470_F00205	32	■ 0.934	$\mathrm{d}_s A$	$\mathrm{d}_s A$	and d_s (space), where $\mathrm{d} = \mathrm{d}_t + \mathrm{d}_s$. Since A contains no dt terms, $\mathrm{d}_s A$ is a 2-form
0707.4470_F00206	32	■ 1.000	$\mathrm{d}_t A$	$\mathrm{d}_t A$	containing only the space terms of dA , while $\mathrm{d}_t A$ contains the terms involving
0707.4470_F00207	32	■ 0.999	$\mathrm{d}_s D = \rho$	$\mathrm{d}_s D = \rho$	law as the sole Euler–Lagrange equation. The divergence constraint $\mathrm{d}_s D = \rho$,

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0707.4470_F00208	32	■ 1.000	<code>t_{f}</code>	t_f	requirement that variations α be fixed at the initial time t_0 and final time t_f . Then,
0707.4470_F00209	32	■ 1.000	<code>\Sigma</code>	Σ	where Σ denotes a Cauchy surface of X , corresponding to the spatial domain. If
0707.4470_F00210	32	■ 1.000	<code>\alpha=\mathrm{d}_{\{s\}} f</code>	$\alpha = \mathrm{d}_s f$	we vary along a gauge transformation $\alpha = \mathrm{d}_s f$, then this becomes
0707.4470_F00211	33	■ 1.000	<code>\mathrm{d}_{\{s\}} D-\rho</code>	$\mathrm{d}_s D - \rho$	This indicates that $\mathrm{d}_s D - \rho$ is a conserved quantity, a momentum map, so if
0707.4470_F00212	33	■ 0.986	<code>\nabla \cdot \mathrm{d}\mathbf{D}</code>	$\nabla \cdot \mathbf{D}$	density $\nabla \cdot \mathbf{D}$ is conserved, but more generally that the <i>change</i> in charge is related
0707.4470_F00213	36	■ 1.000	<code>\mathbb{R}^3</code>	$\mathbb{R}^3$	Nédélec, J.-C. (1980), Mixed finite elements in $\mathbb{R}^3$. <i>Numer. Math.</i> , 35 (3), 315–341.